

Imagining Computing Futures and Mitigating Algorithmic Harm: Conversations Between Artistic Disciplines and Computing

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ABSTRACT

This SIG invites interdisciplinary researchers in ethics, computing, and the arts; data professionals; and arts practitioners into a collective reflection on how these disciplines can work together to mitigate algorithmic harm. As we explore this topic, we recognize the historical and current marginalization of artistic disciplines in terms of credit and funding, and bring into our discussion consideration not only of what computing and data professions can gain from the arts, but of what artistic disciplines can gain from working with computing and data professions guided by the shared goal of mitigating algorithmic harm. This SIG invites reflection on how arts-based methods can help answer the following questions:

- (1) What agency and responsibility do data professionals as individuals have for mitigating algorithmic harm in their day-to-day workflows?
- (2) How does a data professional's individual agency and responsibility relate to collectives (e.g., institutions, groups of colleagues, families, employers, etc.) that they are a part of?
- (3) What agency and responsibility do collectives have for mitigating algorithmic harm?
- (4) What can artistic disciplines gain from working with computing and data professions to mitigate algorithmic harm?

During this SIG, we will introduce the necessary disciplinary contexts for our collective discussion, facilitate a movement improvisation score that we developed called "On The Perils of Poorly Chosen Sorting Algorithms", and facilitate reflective discussion around how arts-based methods can invite data professionals to imagine new data workflows that mitigate algorithmic harm.

CCS CONCEPTS

• **Social and professional topics** → **Codes of ethics**; • **Computing methodologies**;

KEYWORDS

Algorithms, Arts Methods, Computing, Ethics

ACM Reference Format:

Angela Schöpke-Gonzalez, Justin Wyss-Gallifent, Charli Brissey, Steph Jordan, and Libby Hemphill. 2024. Imagining Computing Futures and Mitigating Algorithmic Harm: Conversations Between Artistic Disciplines and Computing. In *Companion of the 2024 Computer-Supported Cooperative Work and Social Computing (CSCW Companion '24)*, November 9–13, 2024, San Jose, Costa Rica. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3678884.3687140>

1 INTRODUCTION

In 2020, a facial recognition algorithm matched a CCTV image of a person committing a crime incorrectly to the identity of a person, ultimately leading to the wrongful detention of Robert Julian-Borchak Williams for a crime he did not commit [17]. In this example of algorithmic harm, what responsibility do researchers have for the harms that Williams incurred? Some scholarship has emphasized the responsibility of datasets used to train algorithms [6, 19, 36]. But, while datasets themselves are important, ultimately researchers and data professionals decide to collect or use those datasets. At the same time, responsibility for decision-making implies that people involved in algorithmic design, implementation, or application (ADIA) have agency to make and change their decisions. Often, researchers' and data professionals' circumstances constrain the amount of agency that they have when they make decisions, and in other cases these individuals may have more agency in decision-making than they imagine.

Notions of agency and responsibility can often be so deeply rooted "that they are taken for granted and even recede from view" [16, p. 20]. However, research has shown that arts-based methods are especially well-suited for surfacing ingrained experiences to consciousness. For six months this year, we (workshop organizers) have collaborated with four other artist-data professionals on a project using arts-based methods movement improvisation [7, 34] and scores [14, 22–24] to surface to consciousness our experiences

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CSCW Companion '24, November 9–13, 2024, San Jose, Costa Rica

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ACM ISBN 979-8-4007-1114-5/24/11

<https://doi.org/10.1145/3678884.3687140>

of agency and responsibility for mitigating algorithmic harms in our day-to-day workflows. Building on our own work and on recent ECSCW workshop “Discomfort in the making of technologies: (re-) choreographing agency” [8], we propose that arts-based methods may support data professionals in understanding their own sense of agency and responsibility for their data work, relative to their contexts.

This SIG invites interdisciplinary researchers in ethics, computing, and the arts; data professionals; and arts practitioners into our collective reflection on how arts-based methods might support data professionals and researchers understanding:

- (1) What agency and responsibility do data professionals as individuals have for mitigating algorithmic harm in our day-to-day workflows?
- (2) How does a data professional’s individual agency and responsibility relate to collectives (e.g., institutions, groups of colleagues, families, employers, etc.) that they are a part of?
- (3) What agency and responsibility do collectives have for mitigating algorithmic harm?

As we discuss what computing and data professions might gain from artistic disciplines, we simultaneously ask at every step, what can artistic disciplines gain from working with computing and data professions to mitigate algorithmic harm?

During this SIG, we will introduce the disciplinary contexts for our collective discussion, facilitate one of the movement improvisation scores that we developed called “On The Perils of Poorly Chosen Sorting Algorithms”, and invite reflective discussion around how arts-based methods can invite us to imagine new data workflows that integrate ethical principles appropriate to our contexts.

2 ETHICAL PRINCIPLES IN DATA WORK

Scholarship over the last decade has been instrumental in articulating the types of harms that algorithms can perpetuate [5, 6, 25, 28]. For example, false positive matches from a facial recognition algorithm used by police have led to the wrongful detention of residents for crimes they did not commit [9, 17]. In another example, for-profit universities used web behavior tracking algorithms to target lower-income people with predatory ads, promising a college education for fees much higher than non-profit colleges. For-profit colleges used these algorithms to lead vulnerable people into tremendous debt, contributing to widening socioeconomic inequality [28]. In response, activists, scholars, and institutions have been articulating principles for how algorithms can actively support just futures rather than passively perpetuate harm. Some examples of principles include those of Data Feminism [10], relational ethics [4], among many others [2, 20, 26].

At an institutional level, governments and organizations have codified some sets of principles into regulations or policies. For example, the European Union’s General Data Protection Regulation (GDPR) [1] codifies ethical principles around data privacy. In the United States, Institutional Review Boards (IRB) codify ethical principles around consent for data collection in their requirements for human subjects research [27]. Alphabet, Google’s parent company, has codified ethical principles that Google employees must abide by in their Code of Conduct [3].

But how do data professionals as individuals within organizations which are required to comply with these ethical principles actually practice these principles in their work? How do these codified principles interact with personally held principles? To what extent do data professionals actually feel agency and responsibility to practice or negotiate ethical principles in their work? Popova et al. [29] recently found that technology practitioners assume responsibility for decision-making to different degrees depending on how much control they perceive they have over the outcomes of their work. Feelings of agency and responsibility are not uniform, but instead highly contingent on context. Given this variability, a one-size-fits-all pathway from principle to practice may not be suitable for data professionals’ lived experiences.

3 FROM PRINCIPLES TO PRACTICE

We propose that reflexive arts-based methods may support data professionals in understanding their own sense of agency and responsibility for their data work, relative to their contexts. Discourse around reflexive methods began emerging in the empirical canon in the 1970s [33], and gained popularity in the 1990s [31] as a way to talk about a researcher’s ability to study how they, as research instruments, affect knowledge production processes and thus the knowledge they produce. Common methods for studying this topic involve reflective writing, including but not limited to keeping a written reflective journal [35], reflective diary [13, 15], research diary [18], or in-process memos [11] throughout a research process. Through writing, researchers can surface conscious and subconscious motivations for decisions throughout their research processes.

Regardless of structure, common to various reflective writing approaches is a grounding in the premise that a researcher is a complex bodied and minded being. Shepherd writes, “Thought and action become integrated; and the reasons, assumptions and feelings that govern behaviour are carefully reflected on... [this process] also enables the writer to capture reflection-in-action, and through this process, provides a means to consciously reflect on tacit practice” [35, p. 334]. However, while the physiological body’s experiences might be understood as central to the researcher’s priorities, these experiences can be difficult to describe [12]. As such, critiques of reflexivity suggest that “There are few attempts to explain how physical actions, whether these be a matter of deliberate behavior or of less-than-conscious habit, produce meaning” [32, p. 333]. This critique invites opportunities for researchers to investigate methods for better surfacing priorities anchored in their physiological bodies.

Concurrently, non-verbal arts-based methods have been shown to surface deeply ingrained subconscious priorities to consciousness in psychotherapeutic contexts. For example, reflecting on a therapist’s use of art therapy methods, Rubin writes that “ideas and feelings would appear in their [patients’] art long before they were expressed in play or verbal therapy” [30, p. 1]. Malchiodi also writes that artistic tasks can support a patient “to communicate what words cannot” [21, p. 2]. In addition to art-making as an expressive form, Malchiodi writes that “the creative process involved in art making, whether it be drawing, painting, sculpting or some

other art form, is what is life enhancing and ultimately therapeutic” [21, p. 2]. These works show that arts-based methods have a documented ability to surface pre-verbal thought to consciousness.

Building on this research, we offer for discussion that reflexive arts-based methods may be helpful for data professionals to understand: (1) their own sense of agency and responsibility for mitigating algorithmic harm in their data work relative to the collectives they are a part of (e.g., institutions, groups of colleagues, families, employers, etc.), (2) how their sense of agency and responsibility might be *constrained* by those collectives, and (3) what agency and responsibility collectives may have for mitigating algorithmic harm.

4 CONCLUSION

This SIG strives to make space for necessary methodologically and epistemologically interdisciplinary conversations around how to move towards equitable, sustainable, and just futures of computing that mitigate algorithmic harm. We highlight complex relationships between individual and collective agency and responsibility for algorithmic harm, and invite discussion around the roles that artistic disciplines can play in understanding those relationships. We anticipate that this SIG will cultivate interdisciplinary community, seed collaborations, and inspire future research on intersections between the arts and computing that can benefit both disciplines.

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